# Long-Context Hallucination Detection in LLM Outputs

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## Abstract

Large Language Models (LLMs) such as ChatGPT have demonstrated remarkable ability to generate fluent, human-like text across a wide range of domains. Despite their strengths, these models are prone to hallucination—producing information that is incorrect, misleading, or entirely fabricated. This problem becomes especially severe when models are required to process long-context inputs or generate citations and references. In this work, we develop a binary classification system to automatically detect whether an LLM-generated response is factual or hallucinated. We evaluate two transformer-based encoder models—DistilBERT and ModernBERT—on HaluEval and LibreEval benchmarks. ModernBERT, with its extended 8,192-token context window, achieves 95.3% accuracy and an F1-score of 0.947 on the combined dataset, substantially outperforming DistilBERT's 87.0% accuracy (F1 = 0.856). Our results confirm that long-context capability is a critical factor in hallucination detection performance.

## 1 Introduction

Large Language Models (LLMs) have achieved impressive results across a diverse set of natural language tasks (Brown et al., 2020). However, a persistent limitation of these systems is their tendency to *hallucinate*—generating outputs that are fluent and confident in appearance yet factually incorrect or entirely fabricated (Ji et al., 2023). This problem is particularly dangerous because hallucinated content is often indistinguishable from truthful content without external verification.

The consequences of hallucinations are significant across many high-stakes domains. In medicine, hallucinated outputs can lead to incorrect diagnoses or unsafe treatment recommendations. In law, fabricated case citations can undermine legal arguments and erode trust in AI-assisted legal research. In education, hallucinated information can mislead students and impair learning outcomes. As deployment of LLMs continues to expand, ensuring the reliability of generated content has become a critical research priority.

The challenge is further compounded when models process long-context inputs—documents, retrieved passages, or multi-turn dialogues that exceed the token limits of standard architectures. Truncating such inputs discards potentially relevant information and introduces systematic blind spots in hallucination detection.

This paper addresses the following central question: *Can transformer-based classifiers reliably detect hallucinations in LLM outputs, and does long-context capability improve detection performance?*

We make the following contributions:

- We fine-tune and evaluate two encoder-only models—DistilBERT and ModernBERT—on two established hallucination benchmarks.
- We demonstrate a clear performance advantage for long-context models, particularly on datasets with extended input sequences.
- We analyze error typologies in hallucinated outputs and discuss computational tradeoffs for deployment.

## 2 Related Work

Hallucination in neural text generation has been studied from multiple angles. Maynez et al. (2020) identified faithfulness failures in abstractive summarization, showing that models often generate content unsupported by source documents. Ji et al. (2023) provided a comprehensive taxonomy of hallucination types in LLMs, distinguishing between intrinsic errors (contradicting the source) and extrinsic errors (introducing unverifiable information).

Benchmark datasets for hallucination evaluation have emerged to support systematic study. HaluEval (Li et al., 2023) provides labeled examples across question answering, dialogue, summarization, and general text tasks, enabling fine-grained evaluation. LibreEval targets more realistic, long-context settings that better reflect production use cases.

Prior detection approaches include natural language inference (NLI) based methods, retrieval-augmented verification, and fine-tuned classifiers (Min et al., 2023). Encoder-only models such as BERT (Devlin et al., 2018) and its derivatives have proven effective for classification tasks requiring cross-sequence reasoning. However, their fixed 512-token context limit poses a fundamental challenge for long-document scenarios.

ModernBERT (Warner et al., 2024) addresses this limitation by extending the context window to 8,192 tokens, making it better suited for detecting inconsistencies in long-form LLM outputs.

# 3 Data

## 3.1 Datasets

This project utilizes two benchmark datasets designed specifically for evaluating hallucination in LLM outputs:

**HaluEval** (Li et al., 2023) comprises 64,507 labeled examples spanning four tasks: question answering (QA), dialogue, summarization, and general text generation. Each example includes a context, an LLM-generated response, and a binary label indicating whether the response is factual (`0`) or hallucinated (`1`).

**LibreEval** adds an additional 48,169 examples organized into non-synthetic, synthetic, and tuning splits. The non-synthetic split reflects naturally occurring text and exhibits very high human label agreement (97.1%), reinforcing the quality of the evaluation targets.

## 3.2 Exploratory Data Analysis

Our exploratory data analysis (EDA) revealed several key characteristics of the datasets. Within hallucinated responses, the most common error types are:

1. **Relation-error**: connecting concepts that should not be related;
2. **Incompleteness**: omitting critical information from the response;
3. **Overclaim**: asserting facts beyond what the context supports.

LibreEval texts are notably long. Reference inputs combined with prompts and outputs frequently exceed the standard 512-token limit. Relying solely on truncation at 512 tokens discards a substantial portion of the available context for many LibreEval samples, motivating the use of a long-context model.

## 3.3 Preprocessing

Before training, data undergoes several preprocessing steps. For each sample, the input text is constructed by combining the original context with the LLM-generated response into a single unified sequence. For question answering, the format is:

```
knowledge [SEP] question [SEP]
answer
```

This formulation allows the model to evaluate the relationship between the source material and the generated output. Text is then tokenized using HuggingFace tokenizers, converting raw text into numerical representations suitable for transformer-based models. Sequence length management differs by model: DistilBERT is restricted to 512 tokens (requiring truncation), while ModernBERT supports sequences up to 8,192 tokens.

The dataset is split into training and evaluation sets using a 90/10 ratio with a fixed random seed. We evaluate models on each dataset independently (HaluEval-only, LibreEval-only) as well as on the combined corpus.

# 4 Models

We compare two transformer-based encoder-only architectures:

**DistilBERT (Baseline)** (Sanh et al., 2019) is a lightweight, distilled version of BERT that retains approximately 97% of BERT's language understanding capability while being 40% smaller and 60% faster. Its maximum input length is 512 tokens, making it efficient for standard-length inputs but limited for long documents.

**ModernBERT (Primary Model)** (Warner et al., 2024) is designed to handle long-context inputs with a maximum sequence length of 8,192 tokens.

| Dataset | Model | Acc. | F1 | AUROC |
|---|---|---|---|---|
| 2*Combined | DistilBERT | 87.0% | 0.856 | 0.959 |
| | ModernBERT | **95.3%** | **0.947** | **0.990** |
| 2*HaluEval | DistilBERT | 80.9% | 0.823 | — |
| | ModernBERT | **93.4%** | **0.932** | — |
| 2*LibreEval | DistilBERT | 96.8% | 0.961 | — |
| | ModernBERT | **98.9%** | **0.986** | **0.996** |

Table 1: Classification performance across datasets. Best results in **bold**.

This extended context window enables ModernBERT to process complete documents without truncation, allowing it to identify subtle inconsistencies between extended source contexts and generated responses.

Both models are fine-tuned as binary classifiers:

$$0 \rightarrow \text{factual response}$$
$$1 \rightarrow \text{hallucinated response}$$

### 4.1 Training Details

Both models are fine-tuned using supervised learning with the following configuration:

- **Learning rate scheduling**: cosine decay to ensure smooth convergence;
- **Dropout**: applied for regularization to reduce overfitting;
- **Early stopping**: training halts when validation performance stops improving (best epochs typically 2–4 for ModernBERT);
- **Mixed precision training**: FP16 used to accelerate GPU computation while maintaining numerical stability.

All experiments are conducted on a GPU cluster (iTiger HPC environment) using single NVIDIA RTX 6000 Ada and H100 nodes.

## 5 Results

### 5.1 Overall Performance

Table 1 presents classification performance across all dataset conditions. ModernBERT consistently outperforms DistilBERT by a substantial margin, confirming the advantage of long-context processing.

| Metric | ModernBERT | DistilBERT |
|---|---|---|
| Training time | ∼4.3 hours | ∼9 minutes |
| Eval time | ∼118 s | ∼5 s |
| F1 | 0.932 | 0.823 |
| AUROC | 0.986 | 0.915 |

Table 2: Speed vs. performance tradeoff on HaluEval-only training. ModernBERT achieves a higher F1 and AUROC but at a dramatically higher computational cost.

### 5.2 Dataset-Specific Analysis

Performance was notably dataset-dependent. On HaluEval, both models performed worst, suggesting that the short, diverse tasks in that benchmark pose unique challenges. DistilBERT's precision on HaluEval dropped to 0.736, indicating a high rate of false positives likely attributable to truncation artifacts. On LibreEval, both models performed well, though ModernBERT's advantage (98.9% vs. 96.8%) demonstrates the continued benefit of full-context processing even when most samples are within manageable lengths.

### 5.3 Computational Tradeoffs

The performance gains of ModernBERT come at a substantial computational cost. Table 2 reports training and evaluation times on HaluEval-only, where the difference is most pronounced. DistilBERT completed training in approximately **9 minutes** with evaluation in **5 seconds**, while ModernBERT required approximately **4.3 hours** of training and **118 seconds** for evaluation—a roughly **29× slower** training time and **24× slower** evaluation time.

Beyond training, inference throughput also differs greatly: approximately **89 samples/second** for ModernBERT compared to **4,165 samples/second** for DistilBERT—nearly a 47× difference. This highlights a strict tradeoff between classification accuracy and real-time deployment feasibility. For applications where latency is critical, DistilBERT may be the more practical choice despite its lower accuracy.

## 6 Discussion

Our results support several key conclusions about hallucination detection in LLM outputs.

**Long-context models are critical.** The most significant performance gap appears on HaluEval, where DistilBERT's input truncation likely discards relevant reasoning chains. ModernBERT's ability to process full sequences without loss directly translates to better hallucination identification.

**Joint input representation matters.** Combining the source context and the generated response into a single input sequence allows the model to directly compare the two and detect inconsistencies. This design choice is essential for high classification accuracy.

**F1 is the appropriate primary metric.** Accuracy can be misleading when class distributions differ across synthetic and non-synthetic splits. F1-score, which balances precision and recall, provides a more informative picture of model behavior.

**Relation-error and incompleteness dominate.** The prevalence of relation-error and incompleteness hallucinations (as opposed to direct factual inversions) suggests that future detection methods should be particularly sensitive to missing or misconnected information.

### 6.1 Limitations

Despite strong overall performance, ModernBERT shows signs of overfitting after early epochs, suggesting that further regularization or improved training strategies may be beneficial. Additionally, the high computational overhead limits its practicality in latency-sensitive production systems. Our evaluation is also limited to two benchmarks derived from a fixed set of LLMs; generalization to other model families remains to be verified.

## 7 Conclusion

We presented a binary classification framework for detecting hallucinations in LLM outputs, comparing DistilBERT and ModernBERT across two benchmark datasets. Our experiments demonstrate that long-context capability is a key driver of detection performance, with ModernBERT achieving 95.3% accuracy and an F1 of 0.947 on the combined dataset. The error typology analysis highlights relation-errors and incompleteness as the dominant hallucination categories, pointing toward targeted directions for future model development.

Future work should explore uncertainty estimation to improve prediction reliability, extend evaluation to a broader range of LLM families, and investigate efficient architectures that reduce the computational gap between long-context and lightweight models for real-time deployment.


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## Contributions

This project was completed collaboratively across five modules (M1–M5). Both authors contributed equally in spirit; the module assignments below reflect primary ownership.

**Sergio Mandujano** led three modules: **M1** (Data Pipeline)—including exploratory data analysis, the dataset loader, and the tokenization and preprocessing pipeline for both HaluEval and LibreEval; **M3** (Training Loop)—implementing the HuggingFace `Trainer` with early stopping, mixed-precision training, learning rate scheduling, and per-model configuration files; and **M5** (Experiment Orchestration)—building the `run_experiment.py` CLI entry point and the `compare_models.py` script that generates the final comparison tables and plots.

**Garrett Sueing** led two modules: **M2** (Model Wrappers)—implementing the DistilBERT and ModernBERT sequence-classification wrappers with a unified interface and long-context attention support for ModernBERT; and **M4** (Evaluation)—implementing the metrics suite (accuracy, F1, AU-ROC), the checkpoint evaluator that produces reproducible JSON reports, and the error analysis notebooks bucketing false positives and negatives by input length.

Both authors contributed to writing and revising the final report.

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